

Supplementary Information for:
Physically-Gated Anomaly Detector: Bridging Process-Based
Simulation and Machine Learning for Pre-Season Risk
Assessment

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Abstract

This document provides technical details, comprehensive data specifications, model hyperparameters, and regional forensic analyses supporting the main text. This material ensures the reproducibility of the physically-gated anomaly detector pipeline and provides evidence for the leak-free standard.

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S1 Master Data Table

The operational forecasting pipeline integrates multi-source data streams to bridge the gap between seasonal climate precursors and field-level physiological responses.

Table S1: Comprehensive Data Sources and Specifications

Dataset	Source	Resolution	Role
Yield Data	Thuringian State Inst. / Destatis [Statistisches Bundesamt (Destatis), 2024]	District (Annual)	Target Variable
AgERA5	ECMWF / Copernicus [Service, 2024]	0.1° (Daily)	Historical Weather
SEAS5	ECMWF [Johnson et al., 2019]	1.0° (Monthly)	Seasonal Forecast
ERA5-Land	ECMWF [Muñoz-Sabater et al., 2021]	0.1° (Monthly)	Soil Moisture / Temp
SoilGrids	ISRIC [Hengl et al., 2017]	250m	Static Soil Properties
CropMap	DLR [DLR (German Aerospace Center), 2023]	10m	Spatial Mask
MOD13Q1	NASA LP DAAC [Didan, 2015]	250m (16-day)	Vegetation Anomalies
Teleconnections	NOAA (NAO, SCAND, MEI)	Monthly	Climate Indices

S2 Predictive Algorithms & Hyperparameters

The physically-gated anomaly detector utilizes specialized components configured for operational stability.

S2.1 Model A: The Hybrid Detector (M_{hybrid})

The hybrid detector is an XGBoost model designed to predict yield deviations from the trend. The Target is defined as $\log(y_t/M_{\text{trend}}(t))$, clipped symmetrically at $[-0.6, 0.6]$. The model uses a set of 106 comprehensive Features, including D_{sow} , S_{heat} , and outputs from the physical prior. To prevent overfitting, the Hyperparameters are heavily regularized (`max_depth=5`, `reg_lambda=2.0`, `learning_rate=0.05`).

S2.2 Model B: The Physical Prior (M_{phys})

The physical prior is a climatological ensemble of WOFOST (v7.2) simulations. Its Configuration consists of 30 parallel simulations. The simulations run on Weather Traces from historical weather data spanning 1995–2024. The model Initialization uses the current year’s soil moisture and temperature (as of March 1st). Finally, the Outputs provided are the Expected Physical Yield (Y_{phys}) and the Downside Risk (10th percentile).

S2.3 Asymmetric Threshold Gate

The gate determines the final forecast by comparing the hybrid detector (M_{hybrid}) with the statistical trend (M_{trend}). The system applies a trigger mechanism when the deviation, defined as $\text{Dev}(t) = (M_{\text{hybrid}}(t) - M_{\text{trend}}(t)) / M_{\text{trend}}(t)$, exceeds a magnitude of 0.04.

When triggered, the system applies asymmetric weighting depending on the direction of the anomaly. On the downside (“crash” signal), the system places full trust in the physical model, setting the final forecast to match the hybrid detector’s prediction. On the upside (“bumper” signal), the system applies a conservative 50% trust, blending the hybrid prediction equally with the statistical trend. If the deviation does not exceed the trigger threshold, the system defaults to the “safety mode,” fully relying on the statistical trend.

Formally, the final forecast is:

$$Y_{\text{final}}(t) = \begin{cases} M_{\text{hybrid}}(t) & \text{if } \text{Dev}(t) < -0.04 \\ 0.5 M_{\text{hybrid}}(t) + 0.5 M_{\text{trend}}(t) & \text{if } \text{Dev}(t) > +0.04 \\ M_{\text{trend}}(t) & \text{otherwise.} \end{cases}$$

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